

Pulse check

- Evaluate the candidates of algo learn so far
- Where do we use neural network in our day to day life
- Inspiration of building NN
- Brief understanding of the history of NN.



→ Log sig is a linear classifier



$$\hat{y} = \frac{1}{1 + e^{-y}}$$

eq. of linear

$$y = mx + c \rightarrow \text{linear sig}$$
$$wx + b$$

Can we improve Logistic Regression?

Task 1 Automatic Complex feature creation

Task 2 Adaptive for multi-class Classification

- we don't have to do it
- works well for high-dimensional data.

NN



Youtube → gradient descent algo
↳ scalable

Autoencoders

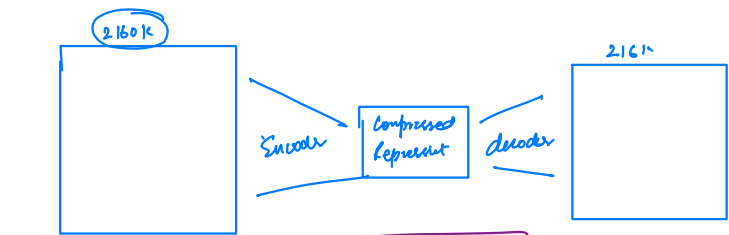
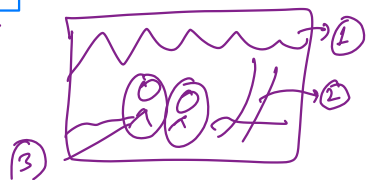
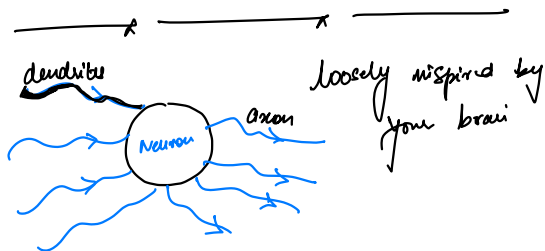


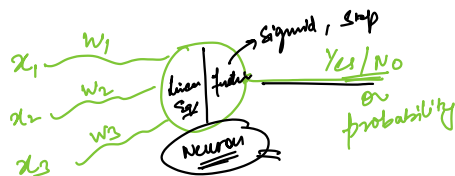
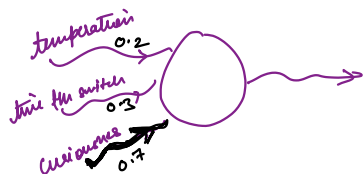
Image Segmentation



Autorecipes, Autoconcepts



lost vision



Sqr

$$\left[\begin{matrix} w_1 x_1 + \\ w_2 x_2 + w_3 x_3 \\ + b \end{matrix} \right] \Rightarrow \text{function}$$

0,1

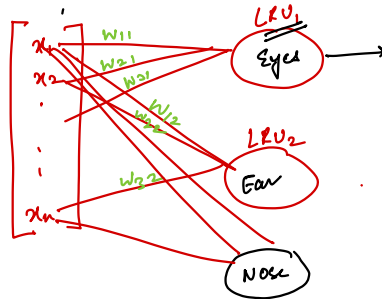
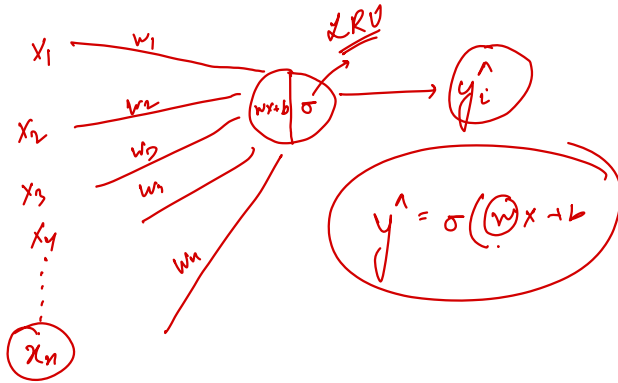
Combination of linear eq. & Sigmoid fcn $\rightarrow$ Log Regression

Neuron = Logistic Regression Unit

Sigmoid

$$Z = (w_1 x_1 + w_2 x_2 + w_3 x_3 + b)$$

= σ



Perceptron → basic neuron → no neural nets

↳ 1957

Backpropagation

- high computational power
- large dataset

} 30 years
1986

1995 - 2009

DT, SVM, GBDT, ..., RF

2012 → Neural networks reemerged.

① On smaller training sets
↳ traditional model